

Question Number	Answer	Additional Guidance	Mark
2 (a)	Any three from: • phospholipid • protein • cholesterol • glycoprotein • glycolipid	The following answers, all 3 correct for 2 marks, 1 or 2 for one mark:	(2)

Question Number	Answer	Additional Guidance	Mark
2 (b)	An answer that includes three of the following points: - the substances are {non-polar / hydrophobic} (1) - there is a positive correlation / as solubility increases so does permeability (1) - because the membrane is less permeable to less hydrophobic substances (1) - therefore membranes contain (phospho)lipids (1)	Accept reverse argument Not just ref to A and E only linear increase ora {fatty acid tails / membranes} are (only) permeable to non-polar substances / non-polar substances move across membrane by dissolving in lipids Accept if implied	(3)

Question Number	Answer	Additional Guidance	Mark
2 (c)(i)	An answer that includes the following points: • increase in {membrane permeability / pigment release / colour intensity} as temperature increases (1) • the change in permeability (between 15 and 20 °C) is due to increased {kinetic energy / movement} of (phospho)lipids (1) • which would cause {phospholipids to move away from each other / a more fluid membrane / a membrane with bigger gaps} (1) • levels off (after 20 °C) because all {pigment / colour} released (1)	DO NOT ACCEPT effect of temperature on transmission of light	(3)

Question Number	Answer	Additional Guidance	Mark
2 (c)(ii)	An answer that includes the following points: • {equal sized / same shaped} pieces of (beetroot) tissue cut (1) • washed in water (until no more pigment lost) (1) • placed in (same / stated) volume of water (1) • placed in a range of temperatures 5 °C to 30 °C (1) • left for {stated / suitable / same} time (1) • samples of the liquid (around the discs) were removed (and placed in colorimeter cuvette) (1) • repeat (at each temperature) {to get mean / SD} (1)	NOT mass on its own 15 mins <=24 hours accept beetroot removed from test tube	(5)